

A Self-Supervised Vision Foundation Model for All-Sky Imagery

Preliminary Validation of a Multi-View Self-Supervised Learning Framework

Yassine Ribouh ¹, Grit Behrens ², Andreas Boschert ³, Mike Zehner ⁴

Abstract: Accurate short-term forecasting of solar irradiance is essential for maintaining the stability of renewable energy grids, yet it remains a difficult task due to the complex and dynamic nature of cloud formations. In this work, we present a vision-based foundation model built using self-supervised learning (SSL) on a large dataset of unlabeled all-sky imagery. Our hybrid SSL framework employs a Vision Transformer (ViT) and integrates masked autoencoding (MAE) to capture fine-grained atmospheric textures, alongside self-distillation to enhance semantic feature learning. To enforce robustness across multiple perspectives, we incorporate a cross-view consistency loss using data from four synchronized camera viewpoints. Early training results show strong convergence, underscoring the effectiveness of our approach as a high-fidelity feature extractor for downstream tasks in solar forecasting.

Keywords: Self-Supervised Learning, Vision Transformer, All-Sky Imagery, Foundation Models, Multi-View Learning, Masked Autoencoding.

1 Introduction

The large-scale integration of solar power is fundamentally challenged by grid instability, a direct result of its inherent intermittency. This variability, driven by dynamic cloud cover, demands accurate short-term irradiance nowcasting to keep the grid reliable. Ground-based all-sky imagers (ASIs) are well suited to the task because they offer finer temporal and spatial resolution than satellites and can capture the local cloud features that

¹ Solar Computing Lab, Bielefeld University of Applied Sciences and Arts, Artilleriestraße 9, 32427 Minden, yassine.ribouh@hsbi.de

² Solar Computing Lab, Bielefeld University of Applied Sciences and Arts, Artilleriestraße 9, 32427 Minden, grit.behrens@hsbi.de,  <https://orcid.org/0009-0009-0247-8204>

³ Laboratory for Solar Energy Systems, Rosenheim Technical University of Applied Sciences, Hochschulstraße 1, 83024 Rosenheim, Germany,  <https://orcid.org/0000-0002-1540-8098>

⁴ Laboratory for Solar Energy Systems, Rosenheim Technical University of Applied Sciences, Hochschulstraße 1, 83024 Rosenheim, Germany,  <https://orcid.org/0000-0001-9642-0496>

matter for plant-level forecasts.

The key problem is turning these high-frequency image streams into physically meaningful feature vectors. Generic computer-vision backbones pre-trained on ImageNet learn edges and object parts that have little to do with atmospheric optics, leaving a semantic gap between the model and the task.

Only a few studies have tried to close this gap with self-supervised learning (SSL) on ASI data. *Johnson et al.* adapted SimCLR to auroral all-sky images and showed that contrastive pre-training boosts aurora-type classification, but the method relies on single-view crops and ignores generative signals [Jo21]. *Fabel et al.* used DeepCluster plus inpainting/super-resolution pre-text tasks on single-camera sky frames to improve four-class cloud segmentation, but they did not exploit multi-camera geometry or joint objectives [Fa22]. Both papers confirm that SSL helps, yet neither tackles view-invariance or combines contrastive and reconstruction losses in one pass.

In this work-in-progress we develop a Vision Transformer (ViT) based foundation model for all-sky imagery. The encoder is pre-trained on a large set of unlabeled sky frames with a hybrid SSL recipe: masked-image reconstruction, DINO-style self-distillation, and a lightweight cross-view consistency loss that ties together synchronized fisheye cameras in our dataset. We hypothesize and early results support that this blend yields a feature extractor that is more attuned to cloud physics than models trained from scratch or with generic pre-training. The remainder of the paper describes the method and reports preliminary gains in downstream nowcasting accuracy.

2 Methodology

2.1 Dataset and Pre-processing

The dataset for this study was collected at a dedicated site in Bittenwiesen, Germany. The setup includes four ASI 16 all-sky imagers from CMS Ing. Dr. Schreder GmbH [Sc25], each paired with a pyranometer and temperature and humidity sensors, enabling synchronized multi-view captures of the sky along with environmental metadata like solar irradiance, air temperature, and relative humidity. Data collection occurred over several discontinuous periods August through October 2024, briefly in November 2024, and again in April and May 2025 interrupted intermittently for system recalibration. Images were originally captured at 15-second intervals



Figure 1: ASI-16/55 with radiation shield in the middle, left the pyranometer and right the temperature and humidity sensor

and then thinned to one-minute spacing to reduce redundancy while retaining meaningful temporal dynamics. For the self-supervised pre-training, the entire dataset was pooled and randomly split, with 98% (146,932 images) used for training and 2% (7,733 images) for validation. While sensor metadata was collected, it was intentionally excluded during SSL training to ensure that representations are learned purely from the image content.

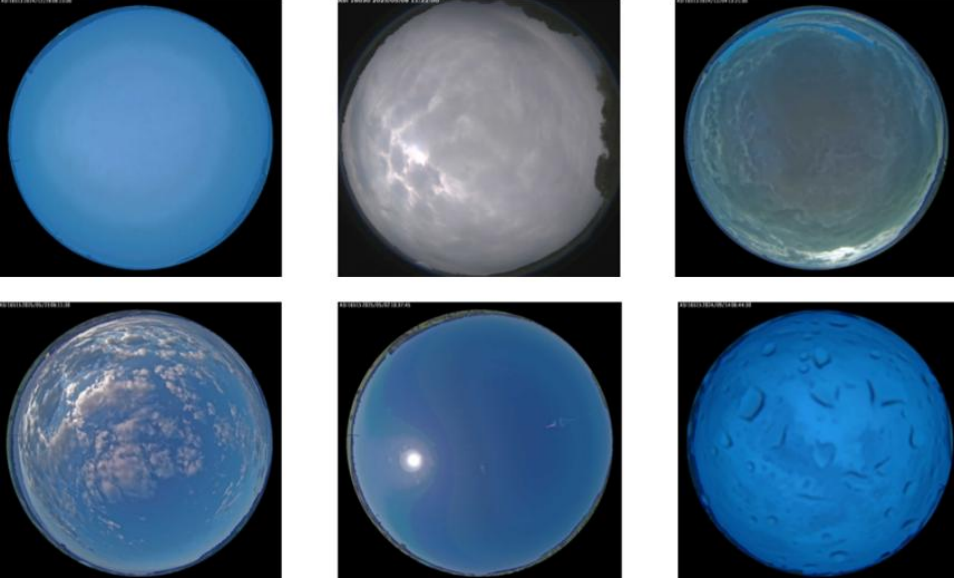


Figure 2: Examples of All-Sky Images with different cloud conditions.

2.2 A Multi-View Hybrid Self-Supervised Architecture

To create a feature extractor that is deeply specialized for the atmospheric domain, we designed a hybrid self-supervised learning framework. This framework is built upon a Vision Transformer (ViT) backbone and combines three distinct but complementary learning objectives: pixel-level reconstruction, high-level feature distillation, and multi-view consistency. The total loss for the model is a weighted sum of the individual loss components:

$$\mathcal{L}_{total} = \lambda_{pixel}\mathcal{L}_{Mae} + \lambda_{distill}\mathcal{L}_{Dino} + \lambda_{xview}\mathcal{L}_{xview}$$

where λ_{pixel} , $\lambda_{distill}$, and λ_{xview} are scalar weights that balance the contribution of each objective.

Backbone: Vision Transformer (ViT)

We selected the Vision Transformer (ViT) architecture as the model's encoder backbone [Do20]. Unlike convolutional neural networks that process images with localized filters, ViT partition an image into a sequence of patches, embed them, and process them using a global self-attention mechanism. This allows the model to capture long-range dependencies across the entire image from the earliest layers. For all-sky imagery, this global context is paramount, as the significance of a cloud in one part of the sky is often dependent on the conditions of the entire sky dome, such as the position of the sun.

Masked Autoencoding (MAE)

The first objective is based on Masked Autoencoding (MAE), a self-supervised method that excels at learning rich, low-level feature representations [He22]. In this approach, a significant portion of the input image patches (e.g., 75%) are randomly masked and removed before being fed to the ViT encoder. The model is then tasked with reconstructing the original pixel values of these missing patches from the latent representation of the visible patches. This task forces the model to develop an internal understanding of the "visual grammar" of the sky learning about typical cloud textures, colour gradients, and the shape of solar glare. The MAE loss is the Mean Squared Error (MSE) computed only on the masked patches:

$$\mathcal{L}_{mae} = \frac{1}{|M|} \sum_{p \in M} ||\text{Decoder}(\text{Encoder}(I_{\text{vis}}))_p - (I_{\text{orig}})_p ||^2$$

where I_{vis} is the original image, I_{orig} are the visible patches, and M is the set of masked patches.

Self-Distillation (DINO-style)

While MAE is powerful for learning local texture, it can be less effective at capturing high-level semantic concepts. To address this, we integrated a self-distillation objective inspired by DINO and DINOv2 (self **D**istillation, **N**O labels) [Ca21, Oq23]. This method uses a student-teacher framework, where the student model (the one being actively trained) learns to match the feature-level output of a teacher model. The teacher model's weights are not updated by backpropagation but are instead an exponential moving average (EMA) of the student's weights from previous training steps.

The student is fed different augmented views of an image and is trained to match the probability distribution produced by the more stable teacher from its own [CLS] (classification) token. This token is prepended to the image patch sequence to act as a summary of the entire image. The objective is formulated as a cross-entropy loss:

$$\mathcal{L}_{\text{Dino}} = - \sum_{x \in V} P_t(x) \log P_s(x)$$

V is the set of augmented views of an input image. $P_s(x)$ is the probability distribution for view x produced by the student network, obtained by applying a temperature-scaled SoftMax function to its [CLS] token embedding. $P_t(x)$ is the target distribution from the teacher network, which is similarly computed but with a different temperature and is sharpened and centered to provide a stable, non-collapsing learning target. This process discourages the model from focusing only on trivial details and pushes it to learn robust, abstract features that are invariant to changes in color, brightness, or orientation.

Multi-View Consistency Loss

The key aspect of our architecture is a cross-view consistency loss designed to explicitly leverage our multi-camera dataset. Because we have four cameras capturing the same scene at the same time from different viewpoints, we can enforce a powerful constraint: the model's high-level understanding of the scene should be independent of the camera from which it was viewed.

To implement this, we augment the input to the ViT. For each image patch sequence, we prepend not only a standard [CLS] token but also a learnable "view embedding" that corresponds to the camera ID (1, 2, 3, or 4). The model is then trained on batches containing simultaneous images from different cameras. The cross-view loss function penalizes the model if the final [CLS] token outputs for these simultaneous views are dissimilar. We use a simple MSE loss for this objective:

$$\mathcal{L}_{xview} = ||z_{cls}^{(i)} - z_{cls}^{(j)}||^2$$

where $z_{cls}^{(i)}$ and $z_{cls}^{(j)}$ are the [CLS] token embeddings for two images of the same scene at the same time, captured from different camera views i and j . This loss forces the model to learn a truly view-invariant representation of the atmosphere, focusing on the inherent physical state rather than camera-specific artifacts or perspectives. This is a critical step towards building a generalized and robust feature extractor.

2.3 Training Protocol

The model is being trained for a total of 400 epochs. As of this writing, training is in progress and has completed 250 epochs on an NVIDIA A100 80 GB GPU, with each epoch taking approximately 20 minutes. We employ the AdamW optimizer [Lo17], configured with a base learning rate of 1.5×10^{-4} and a weight decay of 0.05. The learning rate follows a cosine decay schedule with a linear warmup over the first 10 epochs, allowing the model to stabilize its initial learning before gradually decreasing the rate to converge on a robust solution. Training uses a global batch size of 512 images.

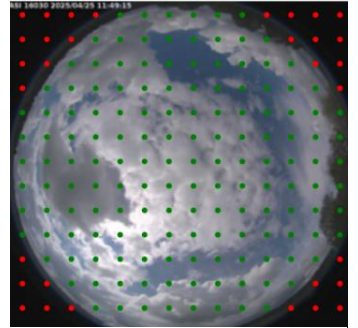


Figure 3: Fisheye mask: valid (green) vs. masked (red) patches

A key implementation detail is the use of a fisheye-aware mask. Since all-sky images have a circular data region surrounded by black, uninformative borders, we apply a static mask during pre-processing. This mask prevents the model from wasting computational resources attempting to reconstruct or learn features from these black pixels, focusing its capacity entirely on the relevant atmospheric data. Furthermore, to stabilize the hybrid learning process, we use a scheduling mechanism for two key hyperparameters. The momentum for the DINO teacher's weight update (an exponential moving average) gradually increased from 0.996 to 0.9995 over the first 30 epochs.

3 Preliminary Results and Analysis

To assess the effectiveness of our self-supervised pre-training strategy at this intermediate stage, we conducted an analysis of the feature space being learned by the ViT encoder.

3.1 Emerging Structure in the Feature Space

We extracted [CLS] embeddings for each validation image and applied t-SNE to map this high-dimensional space into two dimensions. In Figure 4, points colored by withheld solar irradiance form a clear, structured manifold rather than a random scatter. Low-irradiance (purple) and high-irradiance (yellow/green) regions separate smoothly, showing that our SSL training inherently clusters images by atmospheric conditions and captures irradiance-related features.

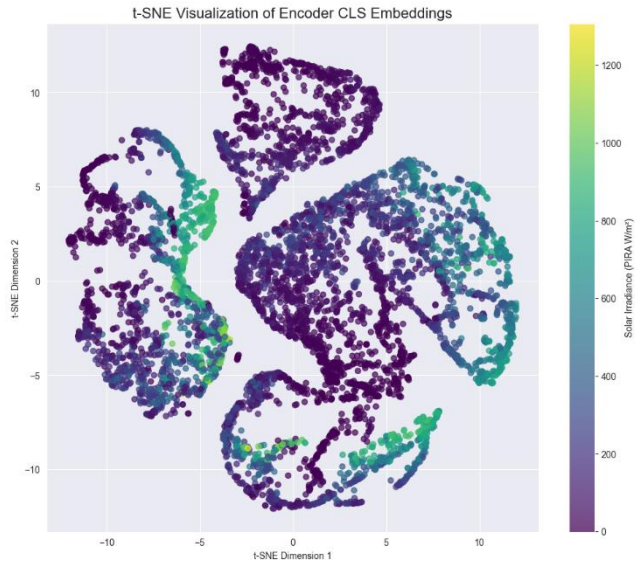


Figure 4: t-SNE Visualization of Encoder Embeddings

We ran K-Means ($k = 3$) on the 2D t-SNE embeddings, yielding three clusters. Sampling images from each cluster then reveals the semantic distinctions among the regions of the learned manifold.

Cluster 1: Overcast Conditions

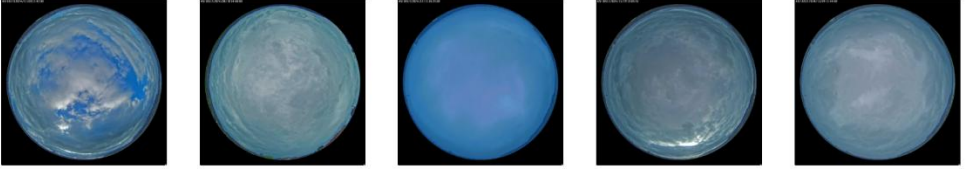


Figure 5: Sample Images from Cluster 1

As shown in **Figure 5**, the first cluster is dominated by images depicting thick, uniform, overcast skies or dense cloud cover.

Cluster 2: Clear Sky Conditions

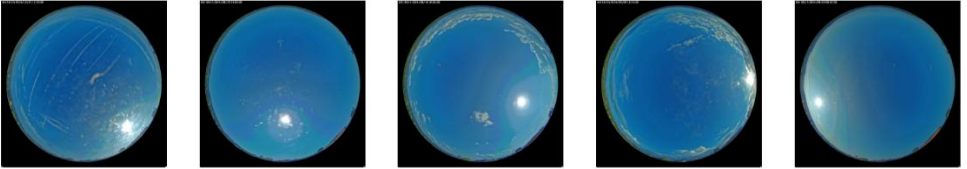


Figure 6: Sample Images from Cluster 2

The second cluster, illustrated in **Figure 6**, contains images of predominantly clear blue skies, sometimes with wispy clouds. The sun is often clearly visible and unobscured.

Cluster 3: Partly Cloudy Conditions

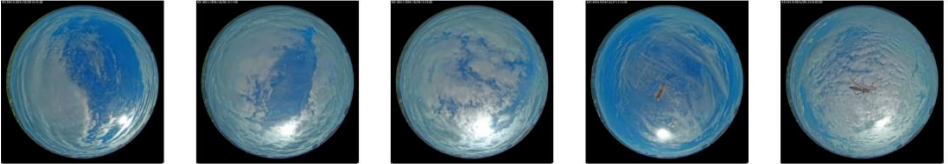


Figure 7: Sample Images from Cluster 3

The final cluster, shown in **Figure 7**, captures the intermediate and most complex conditions. It consists of images with broken clouds, scattered clouds, or hazy skies.

The t-SNE visualization and the subsequent clustering analysis provide initial evidence that our model is successfully learning to disentangle complex atmospheric scenes into a semantically meaningful feature space, confirming our methodology is on the right track.

3.2 Cloud Classification

We assembled a balanced dataset of 1 000 all-sky images, evenly divided among four

classes: **clear sky**, **partial clouds**, **heavy clouds**, and **full overcast**. We randomly split the data into Training (800 images), Validation (200 images), Test (200 images).

We trained the SSL encoder with a lightweight classifier head. Input images were resized to 224×224 and fed in batches of 16, with random horizontal flips and color-jitter augmentations. Each model was trained for up to 60 epochs (early stopping on validation loss) using the cross-entropy loss. As baselines, we evaluated ViT-Base pretrained on ImageNet and a randomly initialized (scratch).

Table 1 reports the per-class F1-scores for cloud classification across the three ViT variants, and our SSL pre-trained model. Our SSL backbone achieves the best overall F1.

Cloud Class	Random F1 (%)	ImageNet F1 (%)	SSL F1 (%)
Clear Sky	78.57	100.00	98.99
Full Overcast	70.69	90.72	93.62
Heavy Clouds	40.91	80.77	82.46
Partial Clouds	38.24	82.83	83.87

Table 1: Per-class F1-scores for cloud classification

4 Discussion and Future Work

4.1 Interpretation of Preliminary Findings

The preliminary results presented in this paper offer strong validation for our hybrid self-supervised learning methodology. The clear, emergent structure in the t-SNE feature space where images cluster by cloud condition demonstrates that even at the halfway point of pre-training, the encoder has learned physically meaningful representations. When we attached a lightweight classifier head and fine-tuned this halfway-trained encoder for cloud classification, it achieved per-class F1-scores of 98.99% (“clear sky”), 93.62% (“full overcast”), 82.46% (“heavy clouds”), and 83.87% (“partial clouds”), outperforming ImageNet and random baselines.

As this is a work in progress, the final performance remains to be determined. Full-schedule pre-training and downstream evaluations will reveal the ultimate efficacy of our approach and guide any necessary tuning of loss weights (λ_{pixel} , λ_{distill} , λ_{xview}) or other hyperparameters. Nonetheless, our halfway-training results already establish a promising foundation for domain-specific sky-image models.

4.2 Roadmap to Completion and Evaluation

Following this assessment, we will employ a larger, more challenging dataset for the **cloud-classification** task building on our initial small-scale experiment to fine-tune the pre-trained encoder. We will also extend fine-tuning to **cloud segmentation** and **solar-irradiance forecasting**, using task-appropriate datasets. Performance on these downstream tasks, benchmarked against models trained from scratch, will deliver a definitive evaluation of our foundation model’s effectiveness.

4.3 Broader Vision and Future Applications

Beyond the immediate goals of cloud classification and irradiance forecasting, the long-term vision for this project is to establish the utility of domain-specific foundation models in the atmospheric sciences. The powerful feature representations learned by our encoder have the potential to serve as a versatile starting point for a range of other important downstream tasks relevant to renewable energy and environmental monitoring. Furthermore, the current model is trained on a dataset capturing several seasons; a logical future step would be to expand the dataset to encompass a full, continuous year of observations. Retraining the model on such a comprehensive dataset would allow it to learn a complete cycle of seasonal variations, further enhancing its robustness and creating an even more powerful tool for atmospheric analysis.

5 Conclusion

In this work-in-progress paper, we have presented the methodology and preliminary validation of a foundation model for all-sky imagery, designed to address the challenge of solar forecasting. Our approach utilizes a hybrid self-supervised learning framework that combines the fine-grained texture learning of masked autoencoding with the high-level semantic feature learning of self-distillation. The key contribution of our architecture is a multi-view consistency loss that leverages simultaneous data from four distinct cameras to enforce a robust, view-invariant scene representation.

Our initial findings are highly encouraging. The analysis of the learned feature space, at an intermediate training stage, demonstrates that the model is successfully organizing complex atmospheric imagery into a coherent manifold that correlates strongly with ground-truth physical properties like cloud conditions. This was achieved without any exposure to explicit labels, providing evidence that our methodology is effectively guiding the model to learn the underlying physics of the atmosphere. This work validates our approach and represents a significant step toward creating specialized, powerful tools for the renewable energy sector. The completed feature extractor is poised to enhance the performance of downstream tasks such as cloud classification and solar nowcasting, ultimately contributing to improved grid stability and the wider adoption of solar power.

Bibliography

- [Ca21] Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., & Bojanowski, P. (2021). Emerging Properties in Self-Supervised Vision Transformers. *arXiv preprint arXiv:2104.14294*.
- [Do20] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., & Houlsby, N. (2020). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. *arXiv preprint arXiv:2010.11929*.
- [Fa22] Fabel, Y., Nouri, B., Wilbert, S., et al. Applying Self-Supervised Learning for Semantic Cloud Segmentation of All-Sky Images. *Atmospheric Measurement Techniques*, 15, 797–809, 2022.
- [He22] He, K., Chen, X., Xie, S., Li, Y., Dollar, P., & Girshick, R. (2022). Masked Autoencoders Are Scalable Vision Learners. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (pp. 15979–15988).
- [Jo21] Johnson, J. W., Hari, S., Hampton, D., Connor, H. K., & Keese, A. A Contrastive Learning Approach to Auroral Identification and Classification. *arXiv:2109.13899*, 2021.
- [Lo17] Loshchilov, I., & Hutter, F. (2017). Decoupled Weight Decay Regularization. *arXiv preprint arXiv:1711.05101*.
- [Oq23] Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., & Bojanowski, P. (2023). DINOv2: Learning Robust Visual Features without Supervision. *arXiv preprint arXiv:2304.07193*.
- [Sc25] Schreder CMS, <http://schreder-cms.com/de/index.htm>, accessed: 18/06/2025.